Yes and no. We have managed to deepen our understanding by breaking free of certain conclusions that, in retrospect, were a consequence of perturbative approximate analyses rather than true string physics. But the current scope of our nonperturbative tools is quite limited. The discovery of the remarkable web of duality relations affords us far greater insight into string theory, but many issues remain unresolved. At present, for example, we do not know how to go beyond the approximate equations for the value of the string coupling constant—equations that, as we have seen, are too coarse to give us any useful information. Nor do we have any greater insight into why there are precisely three extended spatial dimensions, or how to choose the detailed form for the curled-up dimensions. These questions require more sharply honed nonperturbative methods than we currently possess.

What we do have is a far deeper understanding of the logical structure and theoretical reach of string theory. Prior to the realizations summarized in Figure 12.11, the strong coupling behavior of each string theory was a black box, a complete mystery. As on maps of old, the realm of strong coupling was uncharted territory, potentially filled with dragons and sea monsters. But now we see that although the journey to strong coupling may take us through unfamiliar regions of M-theory, it ultimately lands us back in the comfortable surrounds of weak coupling—albeit in the dual language of what was once thought to be a different string theory.

Duality and M-theory unite the five string theories and they suggest an important conclusion. It may well be that there aren’t other surprises, on par with the ones just discussed, that are awaiting our discovery. Once a cartographer can fill in every region on a spherical globe of the earth, the map is done and geographical knowledge is complete. That’s not to say explorations in Antarctica or on an isolated island in Micronesia are without scientific or cultural merit. It only means that the age of geographic discovery is over. The absence of blank spots on the globe ensures this. The “theory map” of Figure 12.11 plays a similar role for string theorists. It covers the entire range of theories that can be reached by setting sail from any one of the five string constructions. Although we are far from a full understanding of the terra incognita of M-theory, there are no blank regions on the map.